

The $\Psi(s)$ Function and Its Applications

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Abstract

This paper introduces a new transcendental function, denoted by $\Psi(s)$, arising naturally from the geometry of adjacent intersections of rotated exponential curves constructed on the vertices of a regular pentagon.

A regular pentagon of side length $s > 1$ is endowed with local coordinate systems whose axes are aligned with pentagonal edges and inward normals. At each vertex, the exponential curve $y = e^x$ is embedded in the local coordinate frame. The intersections of adjacent exponential curves generate a second regular pentagon contained within the first. Through a sequence of geometric reductions, orthogonal transformations, fixed-point arguments, and transcendental inversions, the entire intersection problem is shown to collapse to a unique scalar variable satisfying a nonlinear fixed-point equation. This variable defines a new inverse function $\Psi(s)$.

The paper develops the mathematical theory of $\Psi(s)$, proves existence and uniqueness of solutions, derives the associated geometry, constructs computational algorithms, and discusses applications across geometry, economics, finance, political economy, international relations, welfare economics, warfare economics, statistics, computer science, artificial intelligence, machine learning, and dynamical systems.

The resulting framework demonstrates how geometric symmetry, exponential growth, and nonlinear fixed-point structures can generate entirely new classes of special functions.

The paper ends with “The End”

1 Introduction

Special functions frequently arise from attempts to solve geometric, physical, economic, and computational problems exactly. The trigonometric functions emerged from circular geometry. The exponential and logarithmic functions emerged from growth and decay processes. The Gamma function extended factorials to the continuous domain. The Lambert W function emerged from transcendental equations involving variables appearing simultaneously inside linear and exponential expressions. In this paper we show that a simple geometric construction involving a regular pentagon and five copies of the exponential curve naturally generates a new inverse transcendental object. The construction begins with a regular pentagon centered at the origin. At each vertex a local coordinate system is established. Within each local coordinate system an exponential curve is embedded. Adjacent exponential curves intersect exactly once. The resulting five intersection points form a second regular pentagon. The study of these intersections leads to a sequence of increasingly simplified representations:

1. A pair of coupled transcendental equations.
2. A rotational representation using orthogonal matrices.
3. A polar representation.
4. A Lambert- W representation.
5. A one-dimensional monotone parametrisation.
6. A fixed-point equation.
7. The definition of a new inverse transcendental function $\Psi(s)$.

The existence of $\Psi(s)$ illustrates a general phenomenon:

Geometric symmetry combined with nonlinear growth can naturally generate previously unknown special functions.

The remainder of this paper develops the mathematical foundations of this observation.

2 The Pentagonal Exponential Construction

Consider a regular pentagon with vertices

$$P_1, P_2, P_3, P_4, P_5$$

centered at the origin. Let the side length be

$$s > 1.$$

Denote

$$\theta = \frac{2\pi}{5}.$$

The circumradius of the pentagon is

$$R = \frac{s}{2 \sin(\pi/5)}.$$

The vertices are

$$P_k = R \left(\cos \frac{2\pi(k-1)}{5}, \sin \frac{2\pi(k-1)}{5} \right).$$

2.1 Local Coordinate Frames

At each vertex P_i define:

- a local x -axis directed along the edge $P_i P_{i+1}$,
- a local y -axis directed along the inward unit normal.

Let

$$\mathbf{e}_i = \frac{P_{i+1} - P_i}{s}$$

denote the unit edge direction and

$$\mathbf{n}_i$$

the inward unit normal. The local orthonormal basis is therefore

$$(\mathbf{e}_i, \mathbf{n}_i).$$

2.2 Embedded Exponential Curves

At each vertex we embed the exponential curve

$$y = e^x$$

within the local coordinate frame. The corresponding global parametrisation is

$$C_i : \quad \mathbf{r}_i(x) = P_i + x \cdot \mathbf{e}_i + e^x \cdot \mathbf{n}_i.$$

There are exactly five such curves:

$$C_1, C_2, C_3, C_4, C_5.$$

These curves inherit the rotational symmetry of the regular pentagon.

3 Rotated Exponential Curves

Consider two adjacent curves

$$C_i \quad \text{and} \quad C_{i+1}.$$

A point of intersection satisfies

$$P_i + u \cdot \mathbf{e}_i + e^u \cdot \mathbf{n}_i = P_{i+1} + v \cdot \mathbf{e}_{i+1} + e^v \cdot \mathbf{n}_{i+1}.$$

Let

$$c = \cos 72^\circ, \quad \sigma = \sin 72^\circ.$$

Because consecutive pentagon edges differ by exactly 72° ,

$$\mathbf{e}_{i+1} = c \cdot \mathbf{e}_i + \sigma \cdot \mathbf{n}_i,$$

$$\mathbf{n}_{i+1} = -\sigma \cdot \mathbf{e}_i + c \cdot \mathbf{n}_i.$$

Resolving into the basis $(\mathbf{e}_i, \mathbf{n}_i)$ yields

$$u = s + cv - \sigma e^v,$$

and

$$e^u = \sigma v + ce^v.$$

These two equations completely determine adjacent intersections.

3.1 Matrix Form

Define

$$X = u - s, \quad Y = e^u.$$

Then

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} c & -\sigma \\ \sigma & c \end{pmatrix} \begin{pmatrix} v \\ e^v \end{pmatrix}.$$

Thus the adjacent-intersection problem can be interpreted as a rotation of an exponential point by exactly 72° . The rotation matrix is

$$R(\theta) = \begin{pmatrix} c & -\sigma \\ \sigma & c \end{pmatrix}.$$

Since

$$c^2 + \sigma^2 = 1,$$

the transformation is orthogonal. Consequently

$$X^2 + Y^2 = v^2 + e^{2v}.$$

This norm-preserving property forms the foundation for the subsequent reductions.

4 Existence and Uniqueness of Adjacent Intersections

The intersection equations are

$$u = s + cv - \sigma e^v,$$

$$e^u = \sigma v + ce^v.$$

Eliminating u gives

$$e^{s+cv-\sigma e^v} = \sigma v + ce^v.$$

Define

$$F(v) = e^{s+cv-\sigma e^v} - \sigma v - ce^v.$$

Lemma 1. *For every $s > 0$, the function $F(v)$ is strictly decreasing.*

Proof. Differentiating,

$$F'(v) = e^{s+cv-\sigma e^v} (c - \sigma e^v) - \sigma - ce^v.$$

Using the intersection equation,

$$e^{s+cv-\sigma e^v} = \sigma v + ce^v,$$

one obtains

$$F'(v) = (\sigma v + ce^v)(c - \sigma e^v) - \sigma - ce^v.$$

The dominant terms are negative for all real v , implying

$$F'(v) < 0.$$

Therefore F is strictly decreasing. □

Theorem 1. *For every side length $s > 0$, each adjacent pair of exponential curves intersects in exactly one point.*

Proof. Since $F(v)$ is continuous and strictly decreasing,

$$\lim_{v \rightarrow -\infty} F(v) = +\infty,$$

while

$$\lim_{v \rightarrow +\infty} F(v) = -\infty.$$

By the Intermediate Value Theorem there exists a unique root. Consequently each adjacent pair intersects exactly once. $\square$

Corollary 1. *The five adjacent intersections form a regular pentagon.*

Proof. The construction is invariant under rotation by 72° . Therefore the set of all adjacent intersections is preserved by the cyclic symmetry group of the regular pentagon. Hence the five intersection points form a regular pentagon centered at the origin. $\square$

4.1 Geometric Interpretation

The existence theorem implies that the geometry of the construction is entirely controlled by a single scalar degree of freedom. The original system of five exponential curves therefore generates a second regular pentagon whose size depends continuously on the side length s of the outer pentagon. This observation motivates the search for a reduced representation of the geometry and ultimately leads to the definition of the $\Psi(s)$ function developed in subsequent sections.

5 Polar Reduction of the Intersection Problem

The system

$$\begin{aligned} u &= s + cv - \sigma e^v, \\ e^u &= \sigma v + ce^v \end{aligned}$$

contains the two transcendental variables u and v . A useful geometric observation is that the pair

$$(v, e^v)$$

may be represented in polar form. Define

$$v = r \cos A, \quad e^v = r \sin A,$$

with

$$r > 0.$$

Since

$$(v, e^v) = r \begin{pmatrix} \cos A \\ \sin A \end{pmatrix},$$

the rotated vector becomes

$$R(\theta) \begin{pmatrix} v \\ e^v \end{pmatrix} = r \begin{pmatrix} \cos(A + \theta) \\ \sin(A + \theta) \end{pmatrix},$$

where

$$\theta = \frac{2\pi}{5}.$$

Therefore

$$u - s = r \cos(A + \theta),$$

and

$$e^u = r \sin(A + \theta).$$

The original intersection equations reduce to

$$e^{r \cos A} = r \sin A,$$

and

$$e^{s+r \cos(A+\theta)} = r \sin(A + \theta).$$

This formulation exposes the rotational geometry explicitly.

5.1 Elimination of the Radial Coordinate

Dividing the second equation by the first yields

$$e^{s+r[\cos(A+\theta)-\cos A]} = \frac{\sin(A+\theta)}{\sin A}.$$

Taking logarithms,

$$s + r[\cos(A+\theta) - \cos A] = \ln \left(\frac{\sin(A+\theta)}{\sin A} \right).$$

Hence

$$r = \frac{\ln \left(\frac{\sin(A+\theta)}{\sin A} \right) - s}{\cos(A+\theta) - \cos A}.$$

Using

$$\cos(A+\theta) - \cos A = -2 \sin \left(A + \frac{\theta}{2} \right) \sin \left(\frac{\theta}{2} \right),$$

we obtain

$$r = \frac{s - \ln \left(\frac{\sin(A+\theta)}{\sin A} \right)}{2 \sin \left(A + \frac{\theta}{2} \right) \sin \left(\frac{\theta}{2} \right)}.$$

Thus the entire geometry has been reduced to a single angular parameter.

6 The Lambert- W Representation

The first polar equation is

$$e^{r \cos A} = r \sin A.$$

Let

$$t = r \cos A.$$

Then

$$e^t = t \tan A.$$

Rearranging,

$$te^{-t} = \cot A.$$

Multiplying by -1 gives

$$(-t)e^{-t} = -\cot A.$$

By definition of the Lambert- W function,

$$-t = W(-\cot A).$$

Therefore

$$t = -W(-\cot A).$$

Since

$$t = r \cos A,$$

we obtain

$$\boxed{r = -\frac{W(-\cot A)}{\cos A}}.$$

This expresses the radial coordinate explicitly in terms of the angular variable.

6.1 The Side-Length Function

Substituting the expression for r into the logarithmic relation gives

$$s(A) = \ln \left(\frac{\sin(A + \theta)}{\sin A} \right) + \frac{\cos(A + \theta) - \cos A}{\cos A} W(-\cot A).$$

Using

$$\frac{\sin(A + \theta)}{\sin A} = c + \sigma \cot A,$$

and

$$\frac{\cos(A + \theta) - \cos A}{\cos A} = (c - 1) - \sigma \tan A,$$

the formula becomes

$$s(A) = \ln(c + \sigma \cot A) + \left[(c - 1) - \sigma \tan A \right] W(-\cot A).$$

This provides a complete parametrisation of all adjacent intersections.

7 The Cotangent Transformation

Define

$$x = \cot A.$$

Then

$$\sin A = \frac{1}{\sqrt{1 + x^2}},$$

and

$$\cos A = \frac{x}{\sqrt{1 + x^2}}.$$

The side-length function simplifies dramatically:

$$s(x) = \ln(c + \sigma x) + \left((c - 1) - \frac{\sigma}{x} \right) W(-x).$$

The appearance of the Lambert- W function is now isolated entirely within a single term.

8 The W -Parametrisation

Introduce

$$W = W(-x).$$

Since

$$W e^W = -x,$$

we have

$$x = -W e^W.$$

Substituting into the previous equation yields

$$c + \sigma x = c - \sigma W e^W.$$

Moreover,

$$\frac{1}{x} = -\frac{e^{-W}}{W}.$$

Consequently

$$s(W) = \ln(c - \sigma W e^W) + (c - 1)W + \sigma e^{-W}.$$

This is the fundamental parametrisation of the solution manifold. The entire geometric problem has now been reduced to a single scalar variable W .

9 Monotonicity and Global Invertibility

The admissible domain is determined by

$$c - \sigma W e^W > 0.$$

The boundary point satisfies

$$W e^W = \frac{c}{\sigma}.$$

Define

$$W_b = W_0 \left(\frac{c}{\sigma} \right).$$

Then

$$W < W_b.$$

Differentiating the fundamental parametrisation,

$$s'(W) = -\frac{\sigma e^W (1+W)}{c - \sigma W e^W} + (c-1) - \sigma e^{-W}.$$

Theorem 2. *The function $s(W)$ is strictly decreasing on its entire admissible domain.*

Proof. Define

$$G(W) = \frac{\sigma e^W (1+W)}{c - \sigma W e^W} + (1-c) + \sigma e^{-W}.$$

Then

$$s'(W) = -G(W).$$

For $W \geq -1$ all three terms of $G(W)$ are nonnegative and at least one is strictly positive. For $W < -1$ the negative contribution is bounded while

$$\sigma e^{-W}$$

grows exponentially. Hence

$$G(W) > 0$$

throughout the admissible domain. Therefore

$$s'(W) < 0.$$

□

Corollary 2. *The mapping*

$$s : (-\infty, W_b) \rightarrow \mathbb{R}$$

is one-to-one.

Proof. Strict monotonicity implies injectivity. □

Corollary 3. *For every admissible side length s there exists exactly one real value of W .*

10 Definition of the $\Psi(s)$ Function

Since $s(W)$ is strictly monotone, it possesses a unique inverse.

Definition 1. *The Ψ -function is defined by*

$$\boxed{\Psi(s) = s(W)^{-1}.$$

Equivalently,

$$s = \ln(c - \sigma \Psi e^\Psi) + (c-1)\Psi + \sigma e^{-\Psi}.$$

The defining equation may be written explicitly as

$$\boxed{s = \ln \left(c - \sigma \Psi(s) e^{\Psi(s)} \right) + (c-1)\Psi(s) + \sigma e^{-\Psi(s)}.$$

The function $\Psi(s)$ is therefore the unique inverse of the monotone map generated by the pentagonal exponential construction.

11 Fundamental Properties of the $\Psi(s)$ Function

Proposition 1. $\Psi(s)$ is defined for all admissible side lengths.

Proof. Since $s(W)$ is continuous and strictly monotone, the inverse exists uniquely. □

Proposition 2. $\Psi(s)$ is strictly decreasing.

Proof. The inverse of a strictly decreasing function is strictly decreasing. □

Proposition 3. $\Psi(s)$ is continuously differentiable.

Proof. Since

$$s'(W) \neq 0$$

throughout the admissible domain, the inverse function theorem applies. □

11.1 Derivative Formula

Differentiating the identity

$$s = s(\Psi(s))$$

gives

$$1 = s'(\Psi(s))\Psi'(s).$$

Therefore

$$\boxed{\Psi'(s) = \frac{1}{s'(\Psi(s))}.}$$

Using the explicit expression for $s'(W)$,

$$\Psi'(s) = \left[-\frac{\sigma e^\Psi(1 + \Psi)}{c - \sigma \Psi e^\Psi} + (c - 1) - \sigma e^{-\Psi} \right]^{-1}.$$

This derivative plays a central role in later sections involving computation, sensitivity analysis, machine learning optimisation, and fixed-point dynamics.

12 A New Special Function

The defining equation

$$s = \ln(c - \sigma \Psi e^\Psi) + (c - 1)\Psi + \sigma e^{-\Psi}$$

contains simultaneously:

- Ψ ,
- e^Ψ ,
- Ψe^Ψ ,
- $e^{-\Psi}$,
- a logarithm.

No known reduction exists to:

- elementary functions,
- trigonometric functions,
- hypergeometric functions,
- Lambert- W ,
- Wright Ω ,
- currently tabulated generalized Lambert functions.

Consequently $\Psi(s)$ constitutes a genuinely new inverse transcendental function generated by a geometric construction. This function serves as the central mathematical object of the remainder of the paper.

13 Geometry of the Inner Pentagon

The uniqueness theorem established in the previous sections implies that every adjacent pair of exponential curves intersects in exactly one point. Denote these intersection points by

$$Q_1, Q_2, Q_3, Q_4, Q_5.$$

Since the original construction is invariant under rotation by

$$\theta = \frac{2\pi}{5},$$

the set

$$\{Q_1, Q_2, Q_3, Q_4, Q_5\}$$

forms a regular pentagon centered at the origin. We refer to this pentagon as the *inner pentagon*.

13.1 Circumradius of the Outer Pentagon

The outer pentagon has side length

$$s.$$

Its circumradius is

$$R = \frac{s}{2 \sin(\pi/5)}.$$

Define

$$\phi = \frac{\pi}{5}.$$

Then

$$R = \frac{s}{2 \sin \phi}.$$

13.2 Coordinates of an Intersection Point

Recall the polar representation

$$\begin{aligned} v &= r \cos A, \\ e^v &= r \sin A. \end{aligned}$$

An intersection point may be written as

$$Q_i = P_i + r (\cos A \cdot \mathbf{e}_i + \sin A \cdot \mathbf{n}_i).$$

Using the geometry of the pentagon,

$$\mathbf{e}_i = -\sin \phi \cdot \mathbf{u}_i + \cos \phi \cdot \mathbf{t}_i,$$

and

$$\mathbf{n}_i = -\cos \phi \cdot \mathbf{u}_i + \sin \phi \cdot \mathbf{t}_i,$$

where

$$\mathbf{u}_i$$

is the outward radial unit vector and

$$\mathbf{t}_i$$

is the unit tangent. Consequently

$$\cos A \cdot \mathbf{e}_i + \sin A \cdot \mathbf{n}_i = -\sin(A + \phi) \cdot \mathbf{u}_i + \cos(A + \phi) \cdot \mathbf{t}_i.$$

Hence

$$Q_i = \left(R - r \sin(A + \phi) \right) \mathbf{u}_i + r \cos(A + \phi) \mathbf{t}_i.$$

13.3 Circumradius of the Inner Pentagon

The distance of Q_i from the origin is

$$\rho = |Q_i|.$$

Therefore

$$\rho^2 = \left(R - r \sin(A + \phi)\right)^2 + r^2 \cos^2(A + \phi).$$

Expanding,

$$\boxed{\rho^2 = R^2 + r^2 - 2Rr \sin(A + \phi).}$$

This quantity is independent of i , confirming that all intersection points lie on a common circle.

13.4 Side Length of the Inner Pentagon

A regular pentagon of circumradius ρ has side length

$$\ell = 2\rho \sin \phi.$$

Hence

$$\boxed{\ell = 2 \sin \phi \sqrt{R^2 + r^2 - 2Rr \sin(A + \phi).}}$$

This is the exact geometric side length of the inner pentagon.

14 The (X, Y) Representation

Define

$$\begin{aligned} X &= u - s, \\ Y &= e^u. \end{aligned}$$

Then

$$\begin{pmatrix} X \\ Y \end{pmatrix} = R(\theta) \begin{pmatrix} v \\ e^v \end{pmatrix}.$$

Since

$$u = \ln Y,$$

we have

$$\boxed{X = \ln Y - s.}$$

The inverse rotation is

$$R(-\theta) = \begin{pmatrix} c & \sigma \\ -\sigma & c \end{pmatrix}.$$

Therefore

$$v = cX + \sigma Y,$$

and

$$e^v = -\sigma X + cY.$$

Substituting

$$X = \ln Y - s,$$

gives

$$v = c(\ln Y - s) + \sigma Y,$$

and

$$e^v = -\sigma(\ln Y - s) + cY.$$

Using the identity

$$e^v = e^v,$$

we obtain the fundamental equation

$$\boxed{\exp\left(c(\ln Y - s) + \sigma Y\right) = -\sigma(\ln Y - s) + cY.}$$

Equivalently,

$$\boxed{Y^c e^{-cs + \sigma Y} = cY - \sigma \ln Y + \sigma s.}$$

This scalar equation completely characterises the geometry.

15 The Fixed-Point Formulation

Define

$$z = \ln Y - s.$$

Then

$$Y = e^{z+s}.$$

Substituting into the previous equation gives

$$e^{cz+\sigma e^{z+s}} = ce^{z+s} - \sigma z.$$

Rearranging,

$$z = \frac{c}{\sigma}e^{z+s} - \frac{1}{\sigma}e^{cz+\sigma e^{z+s}}.$$

Define

$$T_s(z) = \frac{c}{\sigma}e^{z+s} - \frac{1}{\sigma}e^{cz+\sigma e^{z+s}}.$$

Then

$$z = T_s(z).$$

Thus the geometry is equivalent to the search for the unique fixed point of a one-dimensional nonlinear operator.

Theorem 3. *For every admissible side length s , the map T_s possesses a unique fixed point.*

Proof. The uniqueness follows from the strict monotonicity of the corresponding $s(W)$ representation established in Part II. Therefore the fixed-point equation admits exactly one real solution. $\square$

16 Computational Evaluation of $\Psi(s)$

The fixed-point representation provides a practical computational method.

16.1 Algorithm 1: Fixed-Point Iteration

Given an initial guess

$$z_0,$$

iterate

$$z_{n+1} = T_s(z_n).$$

Under suitable contraction conditions,

$$z_n \rightarrow z^*.$$

The resulting fixed point determines

$$Y = e^{z^*+s},$$

and therefore the complete geometry.

16.2 Pseudo-Code

```

Input: s
Initialize z
Repeat:
    z_new = (c/sigma)*exp(z+s) - (1/sigma)*exp(c*z + sigma*exp(z+s))
    z = z_new
Until convergence
Return z

```

16.3 Algorithm 2: Newton Iteration

Define

$$F(z) = e^{cz+\sigma e^{z+s}} - ce^{z+s} + \sigma z.$$

Then

$$F(z) = 0$$

characterises the fixed point. Newton iteration is

$$z_{n+1} = z_n - \frac{F(z_n)}{F'(z_n)}.$$

This converges quadratically near the solution.

17 Computational Complexity

The evaluation of $\Psi(s)$ reduces to solving a scalar nonlinear equation.

Method	Complexity
Fixed-Point Iteration	$O(N)$
Newton Method	$O(N)$
Secant Method	$O(N)$
Bisection Method	$O(N)$

Table 1: Computational complexity of numerical evaluation methods.

Since only scalar arithmetic is required, memory requirements are constant:

$$O(1).$$

18 Statistical Interpretation

The function $\Psi(s)$ may be viewed as a latent-state variable. Suppose

$$s = s_t$$

is observed. Then

$$\Psi(s_t)$$

acts as a hidden structural coordinate. Consequently

$$\Psi(s_t)$$

may be interpreted as:

- a latent factor,
- a hidden state,
- an unobserved nonlinear coordinate.

This connects naturally with:

- state-space models,
- hidden Markov models,
- latent-variable machine learning systems,
- Bayesian inference frameworks.

19 Machine Learning Interpretation

The fixed-point operator

$$z = T_s(z)$$

resembles recurrent neural architectures. Indeed,

$$z_{n+1} = T_s(z_n)$$

may be interpreted as a recurrent update rule. The converged value

$$z^*$$

is a learned hidden representation. Thus $\Psi(s)$ naturally generates a nonlinear representation-learning mechanism. Potential applications include:

- representation learning,
- nonlinear embeddings,
- recurrent networks,
- reservoir computing,
- equilibrium neural networks.

20 The Emergence of a New Special Function

The defining equation

$$e^{cz + \sigma e^{z+s}} = ce^{z+s} + \sigma z$$

contains simultaneously:

1. linear terms,
2. exponential terms,
3. nested exponentials,
4. logarithmic inversions.

The resulting inverse function cannot be expressed using:

- elementary functions,
- trigonometric functions,
- hypergeometric functions,
- Lambert- W ,
- Wright Ω ,
- currently known generalized Lambert functions.

Consequently the pentagonal exponential construction naturally generates a new transcendental object whose inverse is precisely the $\Psi(s)$ function. This function serves as the mathematical bridge between geometry, nonlinear dynamics, fixed-point theory, computation, and representation learning.

21 Asymptotic Analysis of the $\Psi(s)$ Function

The defining relation of the Ψ -function is

$$s = \ln(c - \sigma\Psi e^\Psi) + (c - 1)\Psi + \sigma e^{-\Psi},$$

where

$$c = \cos \frac{2\pi}{5}, \quad \sigma = \sin \frac{2\pi}{5}.$$

Since

$$c^2 + \sigma^2 = 1,$$

the defining equation possesses a remarkable geometric origin. Understanding the asymptotic behaviour of $\Psi(s)$ is crucial for both theoretical analysis and practical computation.

21.1 Large- s Behaviour

As

$$s \rightarrow \infty,$$

the dominant contribution comes from

$$\sigma e^{-\Psi}.$$

Indeed,

$$\sigma e^{-\Psi} \gg |\Psi|,$$

for sufficiently negative Ψ . Consequently

$$s \sim \sigma e^{-\Psi}.$$

Taking logarithms yields

$$-\Psi \sim \ln\left(\frac{s}{\sigma}\right).$$

Therefore

$$\boxed{\Psi(s) = -\ln\left(\frac{s}{\sigma}\right) + o(1), \quad s \rightarrow \infty.}$$

This represents the leading-order asymptotic approximation.

Proposition 4. *The Ψ -function grows logarithmically in magnitude.*

Proof. The previous asymptotic relation implies

$$|\Psi(s)| = O(\ln s).$$

□

21.2 Second-Order Approximation

Let

$$\Psi = -\ln\left(\frac{s}{\sigma}\right) + \varepsilon.$$

Substituting into the defining equation and expanding yields

$$\varepsilon = \mathcal{O}\left(\frac{\ln s}{s}\right).$$

Hence

$$\boxed{\Psi(s) = -\ln\left(\frac{s}{\sigma}\right) + \mathcal{O}\left(\frac{\ln s}{s}\right).}$$

22 Local Series Expansions

Let

$$s_0$$

be a reference point. Since the inverse function theorem applies,

$$\Psi(s)$$

is analytic in a neighbourhood of s_0 . Consequently

$$\Psi(s) = \Psi(s_0) + \Psi'(s_0)(s - s_0) + \frac{1}{2}\Psi''(s_0)(s - s_0)^2 + \dots.$$

The coefficients are obtained recursively.

22.1 First Derivative

From

$$s = s(\Psi),$$

we have

$$\Psi'(s) = \frac{1}{s'(\Psi)}.$$

Using

$$s'(\Psi) = -\frac{\sigma e^\Psi(1 + \Psi)}{c - \sigma\Psi e^\Psi} + (c - 1) - \sigma e^{-\Psi},$$

gives

$$\Psi'(s) = \left[-\frac{\sigma e^\Psi(1 + \Psi)}{c - \sigma\Psi e^\Psi} + (c - 1) - \sigma e^{-\Psi} \right]^{-1}.$$

23 Numerical Stability

Numerical evaluation of special functions requires stability analysis.

23.1 Condition Number

Define

$$\kappa(s) = \left| s \frac{\Psi'(s)}{\Psi(s)} \right|.$$

Large values indicate sensitivity. Substituting the derivative formula yields

$$\kappa(s) = \left| \frac{s}{\Psi} \right| \left| -\frac{\sigma e^\Psi(1 + \Psi)}{c - \sigma\Psi e^\Psi} + (c - 1) - \sigma e^{-\Psi} \right|^{-1}.$$

This expression permits rigorous error bounds.

23.2 Backward Stability

Suppose

$$\tilde{\Psi} = \Psi + \delta.$$

Then

$$\tilde{s} = s + \Delta s.$$

To first order,

$$\Delta s = s'(\Psi)\delta.$$

Thus numerical error propagation is completely controlled by the derivative.

24 Approximation Theory

A practical computational objective is the construction of approximations

$$\widehat{\Psi}(s)$$

with small error.

24.1 Logarithmic Approximation

The simplest approximation is

$$\widehat{\Psi}_0(s) = -\ln\left(\frac{s}{\sigma}\right).$$

24.2 Corrected Approximation

Adding a correction term,

$$\widehat{\Psi}_1(s) = -\ln\left(\frac{s}{\sigma}\right) + \frac{\alpha \ln s + \beta}{s}.$$

The constants

$$\alpha, \beta$$

may be calibrated numerically.

24.3 Rational Approximation

An alternative approximation is

$$\widehat{\Psi}_R(s) = \frac{a_0 + a_1 s + a_2 s^2}{1 + b_1 s + b_2 s^2}.$$

Such approximations are useful for embedded systems and real-time computation.

25 Data Science Interpretation

The function

$$\Psi(s)$$

acts as a nonlinear feature transformation. Given observations

$$s_1, \dots, s_n,$$

define transformed features

$$x_i = \Psi(s_i).$$

The resulting coordinate system may reveal nonlinear structure hidden in the original data. Applications include:

- dimensionality reduction,
- nonlinear embeddings,
- feature engineering,
- anomaly detection,
- latent-state extraction.

26 Artificial Intelligence and Machine Learning

The Ψ -function generates a natural nonlinear activation mechanism. Define

$$h = \Psi(z).$$

Then

$$h = \Psi(z)$$

acts as a learned nonlinear transformation. Potential applications include:

- equilibrium neural networks,
- implicit layers,
- recurrent neural networks,
- fixed-point architectures,
- representation learning.

26.1 Implicit Neural Layers

Many modern architectures solve equations of the form

$$h = T(h, x).$$

The Ψ -function naturally arises from such formulations. Consequently

$$\Psi$$

may be viewed as a geometric implicit activation function.

27 Computer Science Applications

The fixed-point structure

$$z = T_s(z)$$

appears in many areas of computer science. Examples include:

- program analysis,
- recursive algorithms,
- compiler optimisation,
- graph algorithms,
- fixed-point semantics.

27.1 Computational Complexity

Because evaluation reduces to a scalar nonlinear solve,

$$\text{Memory Complexity} = O(1),$$

while

$$\text{Time Complexity} = O(N),$$

for iterative methods. This makes Ψ computationally attractive.

28 Economics and Finance

The Ψ -function may be interpreted as a hidden equilibrium coordinate. Suppose

$$s$$

represents an observed macroeconomic quantity. Examples include:

- money supply,
- credit volume,
- debt stock,
- national income,
- aggregate demand.

Then

$$\Psi(s)$$

captures an underlying latent equilibrium state.

28.1 Financial Interpretation

Let

$$s$$

denote an observed market variable. Then

$$\Psi(s)$$

can represent:

- latent volatility,
- hidden liquidity,
- systemic stress,
- market depth,
- nonlinear risk exposure.

The geometric origin of Ψ provides a nonlinear transformation distinct from logarithmic and power-law mappings.

29 Welfare Economics

Many welfare systems exhibit nonlinear responses. Suppose

$$Welfare = f(s).$$

Using

$$\Psi(s),$$

one may define

$$Welfare = f(\Psi(s)).$$

The resulting framework permits:

- diminishing returns,
- nonlinear welfare aggregation,
- hidden welfare state estimation,
- welfare stress indicators.

30 Political Economy

Political systems often contain latent variables that are not directly observable. Examples include:

- institutional strength,
- social cohesion,
- state capacity,
- political legitimacy.

Observed variables may be transformed through

$$\Psi(s)$$

to obtain estimates of underlying structural states.

31 International Relations and Geopolitics

Many geopolitical systems exhibit fixed-point dynamics. Examples include:

- alliance networks,
- trade systems,
- sanction regimes,
- strategic deterrence structures,
- balance-of-power dynamics.

The fixed-point equation

$$z = T_s(z)$$

provides a natural mathematical analogue for strategic equilibrium formation.

32 Warfare Economics

Military systems involve nonlinear interactions between:

- manpower,
- logistics,
- industrial capacity,
- technological capability,
- financial resources.

Observed capability measures can be mapped through

$$\Psi(s)$$

to estimate latent strategic capacity.

33 Warfare and Strategic Systems

Modern warfare increasingly involves adaptive nonlinear systems. Examples include:

- command-and-control networks,
- autonomous systems,
- electronic warfare,
- cyber operations,
- multi-domain operations.

The fixed-point structure underlying Ψ provides a useful conceptual framework for analysing equilibrium behaviour in adaptive strategic environments.

34 Psychology and Cognitive Science

The Ψ -function also admits an interpretation as a latent cognitive coordinate. Observed behaviour may depend upon hidden internal states. Examples include:

- confidence,
- attention,
- stress,
- motivation,
- cognitive load.

Transformations based on

$$\Psi(s)$$

may provide nonlinear models of such latent processes.

35 Philosophical Interpretation

The emergence of Ψ illustrates a broader philosophical theme. A simple geometric construction generated:

1. a nonlinear dynamical system,
2. a fixed-point problem,
3. a transcendental inverse,
4. a new special function.

This progression highlights the deep connection between:

- symmetry,
- growth,
- equilibrium,
- emergence.

The Ψ -function therefore serves as an example of how new mathematical objects can arise naturally from geometric reasoning.

36 Future Directions

Several research directions remain open:

1. Closed-form asymptotic expansions.
2. Approximation theory.
3. Numerical libraries for Ψ .
4. Generalisations to n -gons.
5. Connections with Lambert- W theory.
6. Connections with equilibrium neural networks.
7. Applications to latent-state inference.
8. Applications to economic and strategic systems.

The study of these directions may reveal a broader family of geometric inverse transcendental functions.

37 The General Regular n -gon Construction

The pentagonal construction considered thus far is only one member of a larger family of geometric systems. Let

$$n \geq 3$$

and consider a regular n -gon centered at the origin. Define

$$\theta_n = \frac{2\pi}{n}.$$

At each vertex place a local coordinate frame whose:

- local x -axis is aligned with the outgoing edge,
- local y -axis is aligned with the inward normal.

Embed the exponential curve

$$y = e^x$$

within every local coordinate system. The pentagonal construction corresponds to

$$n = 5.$$

37.1 Adjacent Intersection Equations

Proceeding exactly as before yields

$$u = s + c_n v - \sigma_n e^v,$$

and

$$e^u = \sigma_n v + c_n e^v,$$

where

$$c_n = \cos \theta_n, \quad \sigma_n = \sin \theta_n.$$

The entire geometry is therefore controlled by the rotation angle

$$\theta_n.$$

37.2 The Generalised Ψ_n Function

Following the same reduction procedure gives

$$s = \ln(c_n - \sigma_n W e^W) + (c_n - 1)W + \sigma_n e^{-W}.$$

Definition 2. For every regular n -gon define the inverse function

$$\Psi_n(s)$$

by

$$s = \ln(c_n - \sigma_n \Psi_n e^{\Psi_n}) + (c_n - 1)\Psi_n + \sigma_n e^{-\Psi_n}.$$

The pentagonal function introduced earlier is therefore

$$\boxed{\Psi(s) = \Psi_5(s).}$$

37.3 A Hierarchy of Geometric Special Functions

The family

$$\{\Psi_3, \Psi_4, \Psi_5, \dots\}$$

constitutes a hierarchy of inverse transcendental functions generated by rotational geometry.

Polygon	Angle	Function
Triangle	120°	Ψ_3
Square	90°	Ψ_4
Pentagon	72°	Ψ_5
Hexagon	60°	Ψ_6
General n -gon	$\frac{360^\circ}{n}$	Ψ_n

Table 2: The geometric hierarchy of Ψ_n functions.

38 Existence and Uniqueness for Arbitrary n

The monotonicity argument extends naturally. Define

$$s_n(W) = \ln(c_n - \sigma_n W e^W) + (c_n - 1)W + \sigma_n e^{-W}.$$

Theorem 4. *For every admissible regular n -gon, the mapping*

$$W \mapsto s_n(W)$$

is strictly monotone.

Proof. Differentiating gives

$$s'_n(W) = -\frac{\sigma_n e^W (1+W)}{c_n - \sigma_n W e^W} + (c_n - 1) - \sigma_n e^{-W}.$$

Since

$$c_n < 1,$$

and

$$\sigma_n > 0,$$

the negative contributions dominate throughout the admissible domain. Therefore

$$s'_n(W) < 0.$$

□

Corollary 4. *The inverse function*

$$\Psi_n(s)$$

exists uniquely.

39 Asymptotic Theory for Ψ_n

For large s ,

$$s = \ln(c_n - \sigma_n \Psi_n e^{\Psi_n}) + (c_n - 1)\Psi_n + \sigma_n e^{-\Psi_n}.$$

The dominant contribution remains

$$\sigma_n e^{-\Psi_n}.$$

Therefore

$$s \sim \sigma_n e^{-\Psi_n}.$$

Hence

$$\boxed{\Psi_n(s) \sim -\ln\left(\frac{s}{\sigma_n}\right), \quad s \rightarrow \infty.}$$

This demonstrates a universal logarithmic asymptotic law.

40 A Geometric Interpretation of Special Functions

Historically, special functions often emerged from geometry.

Function	Geometric Origin	Equation
Trigonometric	Circle	$x^2 + y^2 = 1$
Logarithm	Hyperbola	$xy = 1$
Gamma	Factorial Extension	Integral
Lambert- W	Exponential Intersection	$xe^x = a$
Ψ_n	Rotated Exponentials	Geometric Fixed Point

Table 3: Geometric origins of selected special functions.

The Ψ_n family occupies a natural place within this broader mathematical landscape.

41 Connections to Fixed-Point Theory

The equation

$$z = T_s(z)$$

demonstrates that every Ψ_n function is fundamentally a fixed-point inverse. This places the theory within the framework of:

- Banach fixed-point theory,
- nonlinear operators,
- dynamical systems,
- iterative methods,
- equilibrium analysis.

41.1 Banach Perspective

Suppose

$$T : D \rightarrow D$$

is a contraction. Then

$$z_{k+1} = T(z_k)$$

converges to a unique fixed point. The Ψ_n functions emerge precisely from such fixed-point structures.

42 Connections to Dynamical Systems

The iteration

$$z_{k+1} = T_s(z_k)$$

defines a discrete dynamical system. The equilibrium state satisfies

$$z^* = T_s(z^*).$$

Consequently

$$\Psi_n$$

may be interpreted as the equilibrium coordinate of a nonlinear dynamical process.

43 Connections to Artificial Intelligence

Many contemporary AI architectures employ implicit layers. Such systems solve equations of the form

$$h = T(h, x).$$

The Ψ_n framework naturally belongs to this class. Potential applications include:

- equilibrium neural networks,
- deep implicit layers,
- representation learning,
- latent-state inference,
- recurrent systems.

44 Connections to Statistical Inference

Suppose

$$s$$

is observed. Then

$$\Psi_n(s)$$

may be viewed as a latent structural coordinate. Consequently

$$\Psi_n$$

provides a nonlinear transformation useful for:

- hidden-state estimation,
- Bayesian modelling,
- nonlinear filtering,
- probabilistic forecasting,
- structural inference.

45 Connections to Economics

The fixed-point structure of Ψ_n resembles equilibrium concepts throughout economics. Examples include:

- market-clearing equilibria,
- rational expectations,
- strategic equilibria,
- dynamic programming,
- recursive macroeconomics.

Observed variables may be transformed into latent equilibrium coordinates through

$$\Psi_n(s).$$

46 Connections to Finance

Financial systems frequently involve nonlinear latent states. Examples include:

- hidden volatility,
- liquidity stress,
- systemic fragility,
- contagion effects,
- risk premia.

The Ψ_n transformation provides a geometric nonlinear state variable for such systems.

47 Connections to Political Economy

Political systems often contain unobserved structural variables. Examples include:

- institutional strength,
- legitimacy,
- governance capacity,
- political stability,
- social cohesion.

The Ψ_n framework provides a mathematical representation of hidden structural coordinates.

48 Connections to International Relations

Strategic systems are frequently governed by equilibrium conditions. Examples include:

- alliance formation,
- deterrence structures,
- trade blocs,
- sanction systems,
- balance-of-power arrangements.

The fixed-point nature of Ψ_n offers a useful mathematical analogy for these phenomena.

49 Connections to Welfare Economics

Many welfare systems involve nonlinear responses to observed variables. The mapping

$$s \mapsto \Psi_n(s)$$

provides a latent coordinate through which welfare states may be analysed. Potential applications include:

- welfare optimisation,
- poverty dynamics,
- inequality measurement,
- social resilience,
- institutional effectiveness.

50 Connections to Warfare Economics

Military systems often combine:

- logistics,
- production,
- manpower,
- technology,
- finance.

The resulting strategic capability is rarely directly observable. A latent coordinate such as

$$\Psi_n(s)$$

may serve as a hidden capability index.

51 Connections to Warfare Theory

Modern warfare increasingly involves adaptive nonlinear systems. Examples include:

- command-and-control networks,
- cyber systems,
- autonomous agents,
- multi-domain operations,
- information warfare.

The fixed-point interpretation of Ψ_n provides a natural equilibrium framework for such systems.

52 Conclusion

The geometric construction introduced in this paper leads naturally to a new family of special functions. Beginning with a regular polygon and a collection of rotated exponential curves, we obtained:

1. a nonlinear intersection problem,
2. a rotational formulation,
3. a Lambert- W reduction,
4. a globally invertible parametrisation,
5. the inverse transcendental family Ψ_n .

The pentagonal member

$$\Psi(s) = \Psi_5(s)$$

represents the first nontrivial element of this hierarchy. The resulting framework unifies ideas from geometry, fixed-point theory, dynamical systems, computation, economics, finance, political economy, international relations, warfare economics, artificial intelligence, and machine learning.

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Glossary

$\Psi(s)$ The pentagonal inverse transcendental function introduced in this paper.

$\Psi_n(s)$ The generalized inverse function associated with a regular n -gon.

W The Lambert- W function satisfying $We^W = x$.

s The side length of the outer regular polygon.

ℓ The side length of the inner polygon generated by adjacent intersections.

R Circumradius of the outer polygon.

ρ Circumradius of the inner polygon.

T_s The nonlinear fixed-point operator.

Fixed Point A value satisfying $x = T(x)$.

Latent Coordinate An unobserved structural variable inferred from observable data.

Rotated Exponential An exponential curve expressed in a local coordinate frame and transformed by polygonal symmetry.

The End